Samuel Silverman

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Research Vision

I develop new approaches to discovering, designing, and manufacturing physical systems. By **integrating artificial intelligence with computational mechanics**, my work builds algorithmic foundations for **making automated physical design fast, reliable, and widely accessible**. With applications spanning diverse engineering domains, this research has the potential to optimize high-performance structures, accelerate the discovery of functional materials, and streamline the path to fabrication.

Education

PhD, Computer Science 2021-Present
Boston University, Boston, MA
Advisor: Emily Whiting

MS, Computer Science 2019-2021
Columbia University, New York, NY

BS, Computer Science 2015-2019
University of Massachusetts Amherst, Amherst, MA

Publications

Dylan J. Balter, Colin McMillen, Alec Ewe, Jonathan Thomas, Samuel Silverman, Latitha Parameswaran, Luis Fernando Velásquez-García, Emily Whiting, Steven Patterson, Hilmar Koerner, Keith A. Brown
Giant Flexoelectricity of Additively Manufactured Polylactic Acid
Additive Manufacturing (2026)

Samuel Silverman, Dylan Balter, Keith A. Brown, Emily Whiting
Random-Walk Microstructures for Differentiable Topology Optimization
ACM Symposium on Computational Fabrication (2025)

Samuel Silverman, Kelsey L. Snapp, Keith A. Brown, Emily Whiting
Data-Driven Nonlinear Deformation Design of 3D-Printable Shells
3D Printing and Additive Manufacturing (2025)

Kelsey L. Snapp, Samuel Silverman, Richard Pang, Thomas M. Tiano, Timothy J. Lawton, Emily Whiting, Keith A. Brown
A Physics-Informed Impact Model Refined by Multi-Fidelity Transfer Learning
Extreme Mechanics Letters (2024)

Kelsey L. Snapp, Benjamin Verdier, Aldair E. Gongora, Samuel Silverman, Adedire D. Adesiji, Elise F. Morgan, Timothy J. Lawton, Emily Whiting, Keith A. Brown
Superlative Mechanical Energy Absorbing Efficiency Discovered Through Self-Driving Lab-Human Partnership
Nature Communications (2024)

Vishnu Nair, Jay L. Karp, Samuel Silverman, Mohar Kalra, Hollis Lehv, Faizan Jamil, Brian A. Smith
Navstick: Making Video Games Blind-Accessible via the Ability to Look Around
ACM Symposium on User Interface Software and Technology (2021)

Employment

Charles River Analytics, Cambridge, MA — March 2021 - August 2021
Software Engineer Intern
Sensing, Perception, and Applied Robotics Division

MathWorks, Natick, MA — May 2019 - August 2019
Engineering Development Group Intern

Open-Source Software

Monad
C++ library for homogenization of structured 2D and 3D grids. It provides solvers and material models for computing effective (homogenized) material tensors from heterogeneous microstructures.

rwmicro
C++ library for generating random-walk microstructures—binary patterns created by a periodic random walk.

gcs
Python library for generating 3D meshes of generalized cylindrical shells.

Invited Talks

Random-Walk Microstructures for Differentiable Topology Optimization — July 20, 2026
SIGGRAPH Graphics4Science Workshop
Los Angeles, CA

Data-Driven Nonlinear Deformation Design of 3D-Printable Shells — October 24, 2025
Materials Day 2025
Boston University, Boston, MA

Data-Driven Nonlinear Deformation Design of 3D-Printable Shells — March 27, 2023
Transformative Robotics Lab
Northeastern University, Boston, MA

Data-Driven Nonlinear Deformation Design of 3D-Printable Shells — March 8, 2023
Computational Fabrication Group
University of Washington, Seattle, WA

Service

Reviewer: **npj Computatonal Materials** 2026, **Pacific Graphics** 2024.

Teaching

Boston University, Boston, MA
Teaching Assistant
CAS CS 581: Computational Fabrication — Spring 2024, 2026
CAS CS 320: Concepts of Programming Languages — Fall 2025
CAS CS 411: Software Engineering — Summer 2023

Columbia University, New York, NY
Course Assistant
COMS 4170: User Interface Design — Fall 2020

Awards

Dean's Fellowship, Boston University, Boston, MA — 2021-2022

Press Coverage

The Brink, *3D Printing Robot Uses AI Machine Learning for US Army Research* May 2024